

1
Multiple Answer
1 point
Question

Which of the following statements is true of a Markov decision process or MDP?

- ☒ An MDP is defined for a fully observable, stochastic environment.
- ☒ A solution or policy must specify what the agent should do for all states that it can reach.
- ☐ An MDP has to have a reward function with discounted rewards, since current rewards are always more valuable to the agent than future rewards.
- ☐ A proper policy always exists for an MDP.

2
Multiple Answer
1 point
Question

Select all the following components that the policy iteration algorithm equation takes into account.

- ☒ The probability of entering a state S' from state S after performing action A .
- ☒ The utility associated with a state S .
- ☒ All of the actions A that an agent can take.
- ☐ The maximum reward that can be obtained after being in state S .

3
True or False
1 point
Question

Policy is implicitly updated in value iteration.

☒ True

☐ False

4
True or False
1 point
Question

Compared to value iteration, policy iteration often converges in fewer iterations but is more costly (i.e. greater time complexity) per iteration.

☒ True

☐ False

5
True or False
1 point
Question

The utility function estimate must be completely accurate in order to get an optimal policy?

☒ False

☐ True

6
True or False
1 point
Question

Policy is explicitly updated in value iteration.

☐ True

☒ False

7
Multiple Choice
1 point
Question

What is the principle of MEU?

- ☐ An agent should maximize their utilities.
- ☒ An agent should maximize the weighted average of their utilities.
- ☐ An agent should do what feels good for me and you.

1. Which of the following are required to solve an MDP? Select all that apply. (Author: Peyman)

- a. A reward function
- b. A deterministic transition function
- c. Initial state
- d. Terminal state

▼ Answer and Explanation

Answer: a, c

Explanation: b is incorrect because the transition function is stochastic. d is incorrect because terminal states are optional and not required.

From the lecture:

- To find a solution to an MDP, one must define the following things:
 1. A set of **states** $s \in S$
 2. A set of **actions** $a \in A$
 3. A stochastic **transition function** $T(s, a, s')$
 - Also called a probability model
 - Transitions are Markovian: the probability of arriving in s' only depends on s and not the history of earlier states.
 4. A **reward function** $R(s, a, s')$
 - Also known as utility function.
 - Instead of only giving a reward at the goal states, the agent can receive a reward at each time step, based on its transition from s to s' via action a , hence $R(s, a, s')$.
 - The "rewards" can be positive or negative but are bounded by some maximum value.
 - Because the decision process is **sequential**, we must specify the utility function on a sequence of states and actions.
 - The rewards are additive

2. Which of the following are true about discounting the rewards in an MDP? Select all that apply. (Author: Peyman)

- a. Discounting facilitates convergence of algorithms that are based on Expectimax recurrence
- b. Discounting can help avoid infinite horizons (accumulating infinite rewards in games that last forever)
- c. Additive utility refers to $0 < \gamma < 1$
- d. Discounted utility refers to $\gamma = 1$

▼ Answer and Explanation

Answer: a, b

Explanation: c and d are incorrect because discounted utility refers to $0 < \gamma < 1$. When $\gamma = 1$, there is no discount and the rewards from the environment history simply add up, resulting in additive utility.

Discounting **facilitates convergence** because as we go further in the future, the value of the exponentiated γ (where $\gamma < 1$) makes future reward vanishingly small. The utility function (shown below) is most affected by near rewards rather than late rewards. Thus the search can be bottomed out at a certain depth.

$$U([r_0, r_1, \dots, r_n]) = \sum_{t=0}^n r_t \gamma^t$$

The problem of infinite horizons can be addressed by three possible strategies: 1. **discounting**, which focuses the horizon on shorter term, 2. **depth-limited search** which converts an infinite horizon to a finite one, or 3. **absorbing states**, which guarantee that for every policy, a terminal state will eventually be reached.

3. The expected utility of a q-state depends on which of the following? Select all that apply. (Author: Peyman)

- a. The transition function: $T(s, a, s')$
- b. The reward function: $R(s, a, s')$
- c. The discount γ
- d. The value of the successor state

▼ Answer and Explanation

Answer: a, b, c, d

Explanation:

- $Q^*(s, a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')]$
 - $T(s, a, s')$ is the probability of ending up in s'
 - $R(s, a, s')$ is the reward associated with the action taken to end up in s'
 - $V^*(s')$ is the optimal value of the successor state
 - γ is the discount
- Use the equation below for **policy extraction by values** to determine the **optimal policy for the green cell**.

$$\pi^*(s) = \arg \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')]$$

▼ Answer and Explanation

Answer: $\pi^*(green) = down$

Explanation:

a	$Q^*(s, a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')]$
Left	$0.8(-1 + 2) + 0.1(-1 + 6) + 0.1(-1 - 10) = 0.2$
Up	$0.8(-1 - 10) + 0.1(-1 + 2) + 0.1(-1 + 6) = -8.2$
Right	$0.8(-1 + 6) + 0.1(-1 + 6) + 0.1(-1 - 10) = 3.4$
Down	$0.8(-1 + 6) + 0.1(-1 + 6) + 0.1(-1 + 2) = 4.6$

So the optimal policy $\pi^*(s) = Down$ because that is the action associated with the greatest Q-value.

6. Consider the following MDP: (Author: Peyman)

- **Actions:** move [Left, Right, Up, Down]
- **Probability model:**
 - Moves in the intended direction: 80%
 - Moves perpendicular to intended direction to the left: 10%
 - Moves perpendicular to intended direction to the right: 10%
- **Reward function:** -1 for a move in any direction including bumping into the wall (in which case the agent remains in the same cell).
- **Discount factor** $\gamma = 1$
- The **optimal values for each state** $V^*(s)$ are given below.

5	8
6	-10
2	6

- Use the equation below for **policy extraction by values** to determine the **optimal policy for the green cell**.

$$\pi^*(s) = \arg \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')]$$

▼ Answer and Explanation

Answer: $\pi^*(green) = down$

- . . .

7. Which statements are true regarding value iteration and policy iteration algorithms. Select all that apply. (Author: Peyman)

- a. Value iteration repeats the Bellman updates at every iteration to calculate new values. This is inefficient because policies converge long before values and optimal policies are what we are interested in.
- b. Policy iteration aims to compute the optimal policy directly by picking a arbitrary non-optimal policy and repeatedly alternating between an evaluation step and improvement step.
- c. Unlike value iteration, policy iteration may fail to produce an optimal policy.
- d. Policy extraction is not a part of policy iteration algorithm

▼ Answer and Explanation

Answer: a, b

Explanation: The problems with value iteration algorithm stem from the fact that it repeats the Bellman updates to compute new values at every iteration.

- Problem 1: It's slow – $O(S^2A)$ per iteration
- Problem 2: The "max" at each state rarely changes
- Problem 3: The policy often converges long before the values. So we spend a lot of time computing exact values which have no impact on the optimal policy.

Policy iteration circumvents these problems by aiming to compute the optimal policy directly. It starts by picking a arbitrary non-optimal policy and repeatedly alternates between an **evaluation step** and **improvement step** until the policy can not be further improved (converges).

Option c is incorrect because policy iteration always produces the optimal policy. Value iteration also produces the optimal policy but does so implicitly.

Option d is incorrect because policy extraction (discussed in question 6) is the key component of the "improvement" step of policy iteration algorithm. The improvement step uses the value obtained from the evaluation step and performs a one-step-look-ahead expectimax to generate a new policy. This one-step-look-ahead is the same as policy extraction. If the new policy is different

9. Which statement is correct about the utility function? Select all that apply. (Author: Peyman)

- a. In deterministic environments (e.g. where minimax applies), the utility function is insensitive to monotonic transformations.
- b. In stochastic environments (e.g. where expectimax applies), the utility function is sensitive to monotonic transformations.
- c. If agent's preferences satisfy the 6 axioms of rationality, there exists a utility function that captures the agent's preferences not only for prizes (A, B , etc.) but also for lotteries $L = [p, A; 1 - p, B]$
- d. The utility function is unique.

▼ Answer and Explanation

Answer: a, b, c

Explanation:

- In **deterministic** environments (e.g. minimax), the utility function is **insensitive to monotonic transformations** because the magnitudes of the values don't matter. Only their order matters. In this setting, the utility function is called a **value function** or **ordinal utility function**. **a** is correct.
- In **stochastic environments** (e.g. expectimax), the utility function is **sensitive to monotonic transformations** because the magnitudes of the values matter and are used in computation of expected utility. **b** is correct.
- **c** is correct and is a central theorem of utility theory. If the preferences are constrained by the 6 axioms, this theorem yields $A \succ B \iff U(A) > U(B)$ and $A \sim B \iff U(A) = U(B)$. Furthermore, this not only applies to prizes A and B , but also to lotteries $L = [p, A; 1 - p, B]$ because the expected value of a lottery $[p_1, S_1; \dots; p_n, S_n]$ can be decomposed to the value of its prizes S_i :

$$U([p_1, S_1; \dots; p_n, S_n]) = \sum_i p_i \times U(S_i)$$

- **d** is incorrect. The fact that the above theorem proves the existence of a utility function does not imply its uniqueness. In fact, many valid utility functions $U'(x)$ can be derived from a single $U(x)$ because the behavior is invariant under positive linear transformations:

$$U'(x) = k_1 U(x) + k_2 \text{ where } k_1 > 0$$

10. Which statements are correct. Select all that apply. (Author: Peyman)

- a. A Bellman update refers to one iteration step when solving the recursive Bellman equations by an iterative approach.
- b. Both value iteration and policy iteration represent dynamic programming approaches to solving an MDP.
- c. MDPs can be solved in polynomial time in the number of states and actions and the number of bits required to specify the model.
- d. Linear programming solvers are more efficient than dynamic programming for solving MDPs.

▼ Answer and Explanation

Answer: a, b, c

Explanation:

- **a** is correct. From page 573, Section 17.2 of the textbook : "The iteration step, called a Bellman update ... is assumed to apply simultaneously to all states at each iteration. If we apply the Bellman update infinitely often, we are guaranteed to reach an equilibrium (see "convergence of value iteration" below), in which case the final utility values must be solutions to the Bellman equations."

8. Which statement is correct about the principle of maximum expected utility (MEU)? Select all that apply. (Author: Peyman)

- a. MEU states that a rational agent should choose the action amongst available actions that has the greatest expected utility.
- b. MEU can be expressed as $a^* = \arg \max_a EU(a)$ where $EU(a) = \sum_{s'} P(s') \times U(s')$.
- c. MEU holds even if a utility function $U(s)$ is selected that does not capture the agent's preferences.
- d. In value iteration algorithm, MEU is used to derive the plan $\pi(s)$ from the values of the q-states $Q(s, a)$.
- e. Every rational agent *explicitly* maximizes the utility function as dictated by MEU.

▼ Answer and Explanation

Answer: a, b, d

Explanation: a and b reflect the definition of MEU.

c is incorrect because if the utility function does not capture the agent's preferences, then the agent's behavior may not be rational (may not be aligned with its preferences and goals). In other words rational behavior requires that $a \succ b \iff U(a) > U(b)$. If the agent's preferences comply with the 6 axioms of rationality, then there exists a utility function that will capture the agents preferences.

d is correct because $Q(s, a) = \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V(s)]$ reflects the expected utility of a q-state and the plan $\pi(s)$ is the action that maximizes $Q(s, a)$. Notice the similarity between the following equations:

$$\pi(s) = \arg \max_a Q(s, a)$$

$$a^* = \arg \max_a EU(a)$$

11. In the context of incentivizing/disincentivizing an agent's actions, which of the following are true? Select all that apply. (Author: Jacob)

- a. A living reward of < 1.0 encourages an agent to terminate the game more quickly.
- b. The higher the discount γ , the more future-focused an agent becomes.
- c. The lower the noise δ , the more likely an agent is to take risks.
- d. The lower the discount rate γ , the more valuable the rewards at distant time steps.

▼ Answer and Explanation

Answer: a, b, c

Explanation:

a: (true) the living reward is more of a game-based feature that either rewards or punishes an agent for remaining alive in the game. A living reward under 1.0 encourages an agent to leave the game, while a living reward over 1.0 incentivizes the agent to remain in the game

b: (true) the discount rate γ is similar to the tension between interest rates, inflation, and returns on investment. The a $\gamma < 1.0$ on a future reward makes it less valuable to the agent (i.e. inflation/interest), while > 1.0 causes future rewards to be more valuable, similar to returns earned on an investment.

c: (true) The noise δ is what introduces stochasticity and risk into the action-taking process. The higher the noise, the lower the probability that an agent will successfully complete the directed action and the higher the probability of an undesirable outcome. With $\delta = 0.0$ leads to deterministic outcomes, and encourages the agent to take risks because the probability of a bad outcome is reduced. With $\delta = 0.5$, the agent has a 50-50 chance of successfully completing the directed action, making its life a coin-flip, and thus discouraging risk-taking

d: (false) From the textbook, "A discount factor of γ is equivalent to an interest rate of $(\frac{1}{\gamma}) - 1$." This means that $\gamma = 0.9$ is equivalent to an interest rate of 11.1%, while a $\gamma = 0.1$ would be a 900% interest rate! Note, importantly, that this interest rate is punitive, such as with a credit card. A factor > 1.0 yields a negative number, which can be seen as the rate of return on an investment: the longer the agent is willing to wait to collect the

12. Which statements are correct? Select all that apply. (Author: D.C.)

- a. The stochastic process is a random process that is made up of random variables.
- b. The Markov process is a random process that is made up of random variables.
- c. In a Markov process, only the current value of a random variable is relevant for future prediction.
- d. In a stochastic process, only the current value of a random variable is relevant for future prediction.

▼ Answer and Explanation

Answer: a, b, c

Explanation:

- a. A stochastic process is a sequence of random variables representing the evolution of some system of random values over time.
- b. A Markov process is a type of stochastic process that is made up of random variables. It has a specific property, which is mentioned in option c.
- c. In a Markov process, the future state depends only on the current state and not on the sequence of events that preceded it. This property is known as the Markov property.
- d. A stochastic process, in general, does not have the property where only the current value of a random variable is relevant for future prediction. This is a specific property of the Markov process.

Inference by Enumeration in Bayes' Net

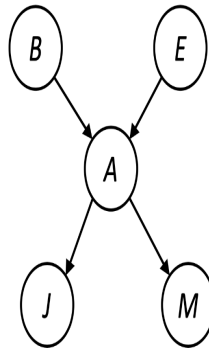
- Given unlimited time, inference in BNs is easy
- Reminder of inference by enumeration by example:

$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

$$= \sum_{e,a} P(B, e, a, +j, +m)$$

$$= \sum_{e,a} P(B)P(e)P(a|B, e)P(+j|a)P(+m|a)$$

$$= P(B)P(+e)P(+a|B, +e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B, +e)P(+j|-a)P(+m|-a) \\ + P(B)P(-e)P(+a|B, -e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B, -e)P(+j|-a)P(+m|-a)$$



13. Which statement is NOT correct? (Author: D.C.)

- a. In finite horizons, policies are typically more complex than in infinite horizons.
- b. Discounted rewards make the utility of an infinite sequence finite.
- c. When the discount factor is close to 0, the agent prefers to wait for long-term rewards.
- d. In infinite horizons, there is no reason to behave differently in the same state at different times.

▼ Answer and Explanation

Answer: c

Explanation: When the discount factor is close to 0, the agent actually prefers to receive immediate rewards and does not want to wait for long-term rewards.

1 Multiple Answer 1 point Question

Which of the following is given to the agent when using passive reinforcement learning? Check all that apply.

- ☒ The agent's policy π .
- ☒ All of the states that can be reached by the agent.
- ☐ The transition model $P(s' | s, a)$.
- ☐ The reward function $R(s)$ that specifies the reward for each state.

2 True or False 1 point Question

Direct utility estimation computes a Utility function U by considering the connection between the utility of a state and the utility of its successor states.

- ☐ True
- ☒ False

7 Multiple Choice 1 point Question

In reinforcement learning, a deterministic policy is

- ☒ A mapping from states to actions
- ☐ A mapping from action to states
- ☐ A mapping from state-action pairs to states

3 Multiple Choice 1 point Question

Which of the following is true of both adaptive dynamic programming (ADP) learning and temporal-difference (TD) learning?

- ☐ A transition model is used in the calculation of the utility of a given state.
- ☒ The utility of a state is adjusted locally to agree with the utility of at least one successor state.
- ☐ Multiple adjustments are made to utilities for each observed transition.
- ☐ A heuristic, such as the prioritized sweeping heuristic, can be used to make adjustments efficiently.

4 Multiple Choice 1 point Question

Which of the following equations are updates for TD Q-learning? Check all that apply.

- ☒ $Q(s, a) \leftarrow Q(s, a) + \alpha(R(s) + \gamma \max_{a'} Q(s', a') - Q(s, a))$
- ☐ $Q(s, a) \leftarrow Q(s, a) + \alpha(R(s) + \gamma Q(s', a') - Q(s, a))$

5 Multiple Choice 1 point Question

What is this equation?

$$U(s) = R(s) + \gamma \max_a \sum_{s'} P(s'|s, a) U(s')$$

- ☐ Temporal difference learning
- ☒ The Bellman equation
- ☐ Policy extraction
- ☐ Policy Iteration

6 True or False 1 point Question

Q-learning with linear function approximation (weighted linear function of a set of features) will always converge to the optimal policy.

- ☐ True
- ☒ False

1. Which is correct? (Author: Peyman)

- a. Solving an MDP involves online planning since all quantities are computed through computation without action.
- b. Reinforcement learning involves offline planning since all quantities are computed through computation without action.
- c. Reinforcement learning is based on a Markov Decision Process for which the probability model and reward function are available.
- d. Reinforcement learning involves online planning since the probability model and reward function for the underlying MDP are not available.

▼ Answer and Explanation

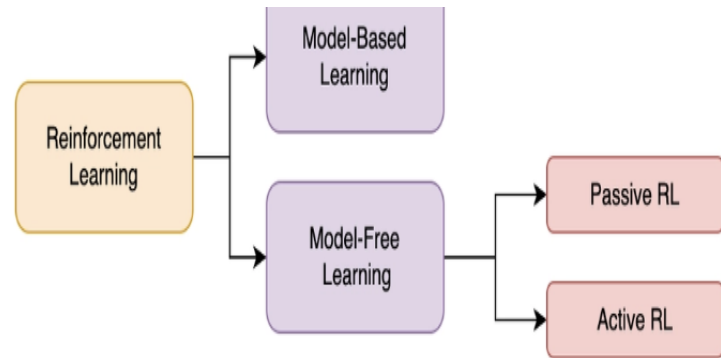
Answer: d

Explanation:

- Solving MDPs involves offline planning:
 - We determine all quantities through computation without action
 - We know the details of the MDP including $T()$ and $R()$
 - We do not take actions to actually play the game!
- Reinforcement learning requires online planning:
 - While there is an underlying MDP, we don't have access to $T()$ and $R()$ so the problem can not be solved by computation alone
 - We must take actions to eventually determine optimal policies and values

2. Which is correct? (Author: Peyman)

- a. Model-based learning is a form of passive reinforcement learning.
- b. Model-based learning has a training step (involving several training episodes) by which the probability model and reward function are estimated.
- c. We have access to or derive the actual probability model and reward function in model-based learning.
- d. The input policy which is used for sampling in the training step of model-based learning is the optimal policy.



3. What is true of passive reinforcement learning? Select all that apply. (Author: Peyman)

- a. Is a form of model-free learning
- b. No policy is provided. The optimal policy is learned.
- c. It eventually computes the correct average values for all states by just using sample transitions.
- d. In passive learning by direct evaluation, because the value of each state is learned independently of other states, it takes it a long time to learn.

▼ Answer and Explanation

Answer: a, c, d

Explanation:

a is correct. Passive reinforcement learning is a form of model free learning. Therefore, it does not compute the probability model or reward function and focuses only on computing the values of all states.

b is incorrect. A policy is provided for passive reinforcement learning. In the simplest form of passive learning, the algorithm works by performing a direct evaluation of this policy. Specifically, in each training episode, it follows the policy from a given state to a terminal state and sums the discounted rewards. It then takes the average of these sums to produce an estimate of the utility of each state.

c and d are correct. Passive reinforcement learning by direct evaluation eventually computes the correct average values for all states by just using sample transitions. However, it takes it a long time (many samples) to learn because it learns the value of each state independently of the other states and does not take into account the connection between states and the structure of the problem.

4. For the same problem as in Question 3, framed as a passive reinforcement learning problem, use direct evaluation to calculate the expected utility (value) of each state, given the same input policy and the observed sample outcomes. Assume $\gamma = 1$. (Author: Peyman)

Training episodes:

1	A, right, B, +5	B, right, C, +7	C, left, B, +5	B, right, C, +7	C, left, T2, +20
2	A, right, T1, -20				
3	B, right, A, +3	A, right, T1, -20			

▼ Answer and Explanation:

In each training episode, we follow the policy from a given state to a terminal state and sum the discounted rewards. We then take the average of these sums to produce an estimate of the utility of each state.

	$V(A)$	$V(B)$	$V(C)$	$V(T1)$	$V(T2)$
Episode 1	$5+7+5+7+20 = 44$	$7+5+7+20 = 39$ $7+20 = 27$	$5+7+20 = 32$ $20 = 20$	—	20
Episode 2	-20	—	—	-20	—
Episode 3	-20	$3-10 = -17$	—	-20	—
Average $V(s)$	$4/3 = 1.33$	$49/3 = 16.33$	$52/2 = 26$	-20	20

6. For a single training episode consisting of the following three transitions and given the initial state values below, use temporal difference learning to update the values for states A, B and C. Assume $\gamma = 0.8$ and $\alpha = 0.5$ (Author: Peyman)

Episode 1	A, right, B, +5	B, right, C, +7	C, left, T2, +20
-----------	-----------------	-----------------	------------------

$V(T1) = 0$		
$V(A) = 2$	$V(B) = 3$	$V(C) = 0$
		$V(T2) = 0$

▼ Answer and Explanation:

Answer: The updated values are:

$V(T1) = 0$		
$V(A) = 4.7$	$V(B) = 5$	$V(C) = 10$
		$V(T2) = 0$

Observed Transition (s, a, s', r)	$V(s)$ initial
A, r, B, 5	2
B, r, C, 7	3
C, l, T2, 20	0

$$V^\pi(s) \leftarrow V^\pi(s) + \alpha(\text{sample} - V^\pi(s))$$

we observed value of the *sample* = $R(s, \pi(s), s') + \gamma V^\pi(s')$

$V(s')$	Observed value of sample $R(s, \pi(s), s') + \gamma V^\pi(s')$	Temporal difference (td) = $\text{sample} - V^\pi(s)$	Updated $V(s)$ = $V(s) + \alpha \cdot td$
3	$5 + 0.8(3) = 7.4$	$7.4 - 2 = 5.4$	$2 + 0.5(5.4) = 4.7$
0	$7 + 0.8(0) = 7$	$7 - 3 = 4$	$3 + 0.5(4) = 5$
0	$20 + 0.8(0) = 20$	$20 - 0 = 20$	$0 + 0.5(20) = 10$

7. What is true of ϵ -greedy exploration? (Author: Peyman)

- The agent acts randomly and explores with a probability $1 - \epsilon$
- The agent follows/exploits current established policy with a probability ϵ
- If a large value for ϵ is selected, then even after learning the optimal policy, the agent will still behave mostly randomly.
- If a large value is selected for ϵ , then the agent will explore infrequently, leading Q-learning (or any other selected learning algorithm) to learn the optimal policy very slowly.

▼ Answer and Explanation

Answer: c

Explanation:

- a** and **b** are incorrect. The opposite is true. ϵ represents the probability of random action:
 - The agent acts randomly and **explores** with **probability ϵ** .
 - The agent follow current established policy and **exploits** with **probability $(1-\epsilon)$** .
- c** is correct. If a large value for ϵ is selected, then even after learning the optimal policy, the agent will still behave mostly randomly.
- d** is incorrect. If a **small** value is selected for ϵ , then the agent will explore infrequently, leading Q-learning (or any other selected learning algorithm) to learn the optimal policy very slowly.
- Conclusion: Because of what was mentioned in c and (the correct version of **d**, it is frequently very difficult to select a fixed value for ϵ and still converge on the optimal policy in a reasonable number of sample observations (as seen in HW7). To get around this, **ϵ must be manually tuned and lowered**

8. What is NOT true of exploration function? (Author: Peyman)

- The agent explores areas whose badness is not yet established and eventually automatically stops exploring.
- The agent has to make a probabilistic decision between exploration and exploitation.
- Exploration is automatically encoded by the exploration function, since the term $\frac{k}{n}$ can give enough of a "bonus" to some infrequently-taken action such that it is selected over actions with higher Q-values.
- As time goes on and states are visited more frequently, the value of $\frac{k}{n}$ decreases towards 0 for each state, effectively ending exploration in favor of exploitation.

▼ Answer and Explanation

Answer: b

Explanation:

9. What is true of both Q-Learning and Approximate Q-Learning? Select all that apply.

(Author: Peyman)

- a. Both are active reinforcement learning paradigms.
- b. Both observe a sample and compute the Q-value of a sample:

$$sample = R(s, a, s') + \gamma \max_{a'} Q(s', a')$$

- c. Both compute the difference between observed and expected Q-values:

$$\Delta = sample - Q(s, a)$$

- d. Both **directly** update $Q(s, a)$ by applying the difference Δ and the learning rate α .

- e. Both always converge on the optimal policy

▼ Answer and Explanation

Answer: a, b, c

Explanation:

d is incorrect because it only applies to Q-Learning (see below). In Approximate Q-Learning, Δ and α are used in combination with feature values to update the feature weights $w_i \leftarrow w_i + \alpha \cdot \Delta \cdot f_i(s, a)$. The Q-value is then computed using the updated feature weights and feature values: $Q(s, a) = \sum_i^n w_i f_i(s, a)$

e is incorrect. Only Q-learning is guaranteed to converge on the optimal policy. Approximate Q-learning uses linear function approximation which is limited in its ability to represent complex state-action Q-value functions and is not guaranteed to converge on an optimal policy. It aims to get a "good-enough" policy.

1 Multiple Answer 1 point Question

Which of the following must be true of the probabilities of two events, A and B, if A is independent of B?

- ☒ $P(A | B) = P(A)$
- ☒ $P(B | A) = P(B)$
- ☒ $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$
- ☒ $P(A \wedge B) = P(A)P(B)$

2 Multiple Answer 1 point Question

Which of the following are equivalent to the joint probability $P(X_1, X_2, X_3, X_4)$?

Check all that apply.

- ☐ $P(X_1)P(X_2 | X_1)P(X_3 | X_2)P(X_4 | X_3)$
- ☒ $P(X_1 | X_2, X_3, X_4)P(X_4)P(X_3 | X_4)P(X_2 | X_3, X_4)$
- ☒ $P(X_2, X_1)P(X_3 | X_2, X_1)P(X_4 | X_3, X_2, X_1)$
- ☒ $P(X_1)P(X_2 | X_1)P(X_3 | X_2, X_1)P(X_4 | X_3, X_2, X_1)$

10. In the context of reinforcement learning, what is true of regret? Select all that apply.

(Author: Peyman)

- a. Regret = optimal rewards - actual rewards accrued
- b. Regret is a measure of total mistake costs during the learning process.
- c. Regret is a measure of how much better we would have done if we had acted optimally throughout the learning process.
- d. ϵ -greedy (random) exploration and exploration function both end up with optimal policies but random exploration has higher regret.

▼ Answer and Explanation

Answer: all are true

3 True or False 1 point Question

In a bigram model, one assumes that words $w[i]$ and $w[i - 2]$ are independent for $i > 2$.

- ☒ True
- ☐ False

4 Multiple Choice 1 point Question

When we compute a tri-gram, we normalize the following to add up to 1:

- ☐ the prior probabilities of all words w
- ☐ the probabilities of all three-length sequences
- ☒ the probabilities of all words w given $w[i - 1], w[i - 2]$

5 Multiple Choice 1 point Question

How can we estimate the probability of a sentence $P(w_1, w_2, \dots, w_N)$?

- ☐ By applying Bayes' Rule.
- ☒ By applying the chain rule.
- ☐ Using a Markov decision process.
- ☐ Using alpha-beta pruning.

Turn

6 Multiple Choice 1 point Question

What is the effect of the Markov assumption in n-gram language models?

- ☒ It makes it possible to estimate the probabilities from data.
- ☐ Long distance relationships, like subject-verb agreement, are taken into account.
- ☐ The probability of a word is determined by all previous words in the sentence.
- ☐ The probability of a word is determined only by a single preceding word.

▼ Module 9 Questions: Probabilities and Language Models

Questions 1 - 6 correspond to the following scenario. In a population of 50 patients, 7 are diabetic and obese, 3 are only diabetic, 15 are only obese, and 25 are neither diabetic nor obese. (Author: Peyman)

1. What is the size of the joint probability distribution for this population if diabetes and obesity are treated as random variables?
- ▼ Answer and Explanation
- Answer: 4
- Explanation:
- Since we are concerned about presence or absence of diabetes and obesity, we can treat them as two binary random variable, each with a domain size of 2. So the size of the joint probability distribution is $d^n = 2^2 = 4$

2. Using random variables D and O , construct a joint probability distribution table for this patient population.
- ▼ Answer and Explanation
- We can use the domain {1,0} for each random variable to represent presence or absence of the condition, respectively. For example $D = 1$ indicates presence of diabetes. The counts in each category are divided by 50 to give normalized probabilities:

D	O	Count	$P(D, O)$
1	1	7	0.14
1	0	3	0.06
0	1	15	0.30
0	0	25	0.50

6. For the two marginal probability distribution tables shown below, select the correct answer.
- a. $p1 = 0.44, p3 = 0.20$
- b. $p1 = 0.20, p3 = 0.44$
- c. $p1 = 0.20, p3 = 0.80$
- d. $p1 = 0.80, p3 = 0.56$

D	$P(D)$
1	$p1$
0	$p2$

O	$P(O)$
1	$p3$
0	$p4$

▼ Answer and Explanation

Answer: b

Explanation:

$$p1 = P(D = 1) = P(D = 1, O = 0) + P(D = 1, O = 1) = 0.06 + P.14 = 0.20$$

$$p3 = P(O = 1) = P(O = 1, D = 0) + P(O = 1, D = 1) = 0.30 + 0.14 = 0.44$$

9. Referring to the table in Question 8, if we have a grade 2 meningioma, what is the probability that we would find a mutation in gene A. Use inference by enumeration. (Author: Peyman)

▼ Answer and Explanation

Answer: 0.58

Explanation:

Step 1. Let G be the evidence variable and A the query variable. Since we are not interested in gene B here, we set it to the hidden variable.

Step 2. Sum the probabilities over the hidden variable B :

$$P(A = 1, G = 2) = \sum_B P(A, B, G) = 0.020 + 0.015 = 0.035$$

Step 3. calculate Z and normalize.

$$Z = \sum_{A,B} P(A, B, G) = 0.020 + 0.015 + 0.015 + 0.010 = 0.060$$

$$P(A = 1|G = 2) = \frac{1}{Z} P(A = 1, G = 2) = \frac{0.035}{0.060} = 0.58$$

3. What is the probability that a patient is diabetic or obese $P(D = 1 \cup O = 1)$?

▼ Answer and Explanation

Answer: 0.50

Explanation:

$$\text{Method 1: } P(D = 1 \cup O = 1) = 1 - P(D = 0 \cap O = 0) = 1 - 0.5 = 0.5$$

$$\text{Method 2: } P(D = 1 \cup O = 1) = P(D = 1) + P(O = 1) - P(D = 1 \cap O = 1)$$

$$= (0.14 + 0.06) + (0.14 + 0.30) - 0.14 = 0.5$$

4. Given that a patient is obese, what is the conditional probability of being diabetic $P(D = 1|O = 1)$?

▼ Answer and Explanation

Answer: 0.32

Explanation:

$$P(D = 1|O = 1) = \frac{P(D=1 \cap O=1)}{P(O=1)} = \frac{0.14}{0.14+0.30} = 0.32$$

5. Give that a patient is not diabetic, what is the conditional probability of being obese $P(O = 1|D = 0)$?

▼ Answer and Explanation

Answer: 0.375

Explanation:

$$P(O = 1|D = 0) = \frac{P(O=1 \cap D=0)}{P(D=0)} = \frac{0.30}{0.30+0.50} = 0.375$$

7. A particular type of brain tumor is called [meningioma](#). Meningiomas are usually benign (grade 1), but about 6% are atypical/invasive (grade 2), and a very small fraction are malignant (grade 3). We are interested in investigating the relationship between mutations in 10 different genes and the grade of meningioma. Consider the 10 gene mutations as random variables $M_1, \dots, M_{10}$. DNA from a large number of surgical tumor specimen of all three grades was assayed for presence ($M_i = 1$) or absence ($M_i = 0$) of each mutation. Making no assumptions about the independence of these random variables, what is the size of the joint probability distribution that considers these mutations and the grade of the tumor? (Author: Peyman)

- a. 2^{10}
- b. 2^{11}
- c. 11^2
- d. $2^{10} \times 3$

▼ Answer and Explanation

Answer: d

Explanation:

8. Grade 2 meningioma are clinically challenging because of their high rate of recurrence after resection. Assume that from our previous analysis we have identified 2 gene mutations, A and B, that may play a role in pathogenesis of grade 2 meningiomas. Representing the presence or absence of a mutation with 1 or 0, we have the following *partial* joint probability distribution that focuses only on grade 2 tumors (grades 1 and 3 are not shown). What is the probability that at least one gene has a mutation in a grade 2 tumor. (Author: Peyman)

A	B	G	$P(A, B, G)$
...	...	...	...
1	1	2	0.020
1	0	2	0.015
0	1	2	0.015
0	0	2	0.010
...	...	...	...

10. Given the following marginal and conditional probability distributions for random variables X_1 and X_2 , produce the joint probability distribution table for $P(X_1, X_2)$. (Author: Peyman)

X_2	$P(X_2)$
1	0.6
0	0.4

X_1	X_2	$P(X_1 X_2)$
1	1	0.3
0	1	0.7
1	0	0.5
0	0	0.5

▼ Answer and Explanation

Answer:

X_1	X_2	$P(X_1, X_2) = P(X_1 X_2) \times P(X_2)$
1	1	0.3 (0.6) = 0.18
0	1	0.7 (0.6) = 0.42
1	0	0.5 (0.4) = 0.20
0	0	0.5 (0.4) = 0.20

Explanation: Use of product rule: marginal probability $P(X_2)$ $\times$ conditional probability $P(X_1|X_2)$ = joint probability $P(X_1, X_2)$. Double check that all probabilities in the joint table add up to 1.

14. In a trigram language model, what is true? (Author: Peyman)

- a. $k = 3$
- b. $k = 2$
- c. We compute the conditional probability of a word given the joint probability of the preceding 3 words.
- d. $P(W) \approx \prod_i^n P(w_i|w_{i-3}, w_{i-2}, w_{i-1})$

▼ Answer and Explanation

Answer: b

Explanation: $k = 2$, so we compute the conditional probability of a word given the joint probability of the preceding 2 words. The applicable equation for a sentence of n words is:

$$P(W) \approx \prod_i^n P(w_i|w_{i-2}, w_{i-1})$$

12. Without any simplifying assumptions, maximum likelihood estimation (MLE) is not a practical technique for estimating the conditional probability of the next word in a sequence of words because: (Author: Peyman)

- a. It does not accurately compute this conditional probability
 - b. Requires a small corpus of sentences
 - c. It is limited by data constraints
 - d. It is not limited by data constraints but is limited by computational constraints because it is inefficient.
- Answer and Explanation

13. In the context of language models, what is true of the Markov assumption? (Author: Peyman)

- a. Similar to its use in MDPs, it assumes that the the probability of arriving at word w_i only depends on the probability of word w_{i-1} and not any of the previous history; i.e., not on any of the preceding words in the sequence.
- b. Is a simplifying assumption that assumes the conditional probability of the next word given a sequence of words only depends on the previous k words, not the entire context.
- c. After applying the Markov assumption as a simplifying assumption, the approximate probability of a sentence becomes:

$$P(W) = P(w_1, w_2, \dots, w_n) \approx \prod_i^k P(w_i|w_{i-1}, \dots, w_{i-k})$$

▼ Answer and Explanation

Answer: b

17. What is true about perplexity? Select all that apply. (Author: Peyman)

- a. The perplexity of a language model on a test set W is the inverse of the joint probability of the test set, normalized by the number of words.
- b. Perplexity is a metric for extrinsic evaluation of a language model
- c. For a sentence W of N tokens, perplexity of a bigram model is represented by the following equation (see HW8). In this equation, normalization is done by taking the $N + 1$ root instead of the N root because of inclusion of the end-of-sentence marker <END>.

$$\text{perplexity}(W) = \sqrt[N+1]{\prod_{i=1}^{N+1} \frac{1}{P(w_i|w_{i-1})}}$$

- d. In order for the perplexity to be accurate, the sequence of tokens W must not be in set of tokens on which the n-gram model was trained.
- e. The higher the perplexity, the better the input sequence (the test set W) is explained by the model.
- f. For a unigram model, a bigram model and a trigram model trained on the same training set and tested on the same test set, the trigram model is likely to have a lowest perplexity.

▼ Answer and Explanation

Answer: a, c, d, f

1 True or False 1 point Question

The conditional probability distribution of a variable in a Bayesian network should be specified based on the probability distributions of all of the other variables (nodes).

- ☐ True
- ☒ False

2 True or False 1 point Question

Every variable in a Bayes net is independent of all of its descendants given its children.

- ☐ True
- ☒ False

3 True or False 1 point Question

Every variable in a Bayes net is independent of all of its descendants given its children.

- ☐ True
- ☒ False

4 Multiple Choice 1 point Question

Consider the Bayes Net image by following this link:

<https://artificial-intelligence-class.org/assets/img/quiz9bayes.png>

Write the joint probability for the Bayes' Net given above, encoding its independence assumptions into your equation.

- ☐ $P(A|D, C) * P(B|E) * P(C|D, E) * P(E) * P(D)$
- ☒ $P(A) * P(B) * P(C|A) * P(E|B, C) * P(D|A, C)$
- ☐ $P(A) * P(B) * P(C|A) * P(E|B) * P(D|A)$
- ☐ $P(A) * P(B) * P(C) * P(D) * P(E)$

5 Multiple Choice 1 point Question

Which of the following is true of locally structured (sparse) systems? Select all that apply.

- ☐ Each subcomponent must interact directly with all the other components.
- ☒ The structure grows linearly in complexity (rather than exponentially).
- ☐ Every variable cannot be influenced by all of the others.
- ☐ All such systems are compact.

6

True or False 1 point Question

More than one Bayesian network can be used to represent the same joint distribution.

- ☒ True
- ☐ False

7

Multiple Choice 1 point Question

If two variables (nodes) X and Y in a Bayesian network do not share a path, which of the following must be true?

- ☐ X and Y can never be true at the same time.
- ☒ X and Y are conditionally independent.
- ☐ X has a direct influence on Y.
- ☐ There exists a causal relationship between X and Y.

Binary Perceptron algorithm examples:

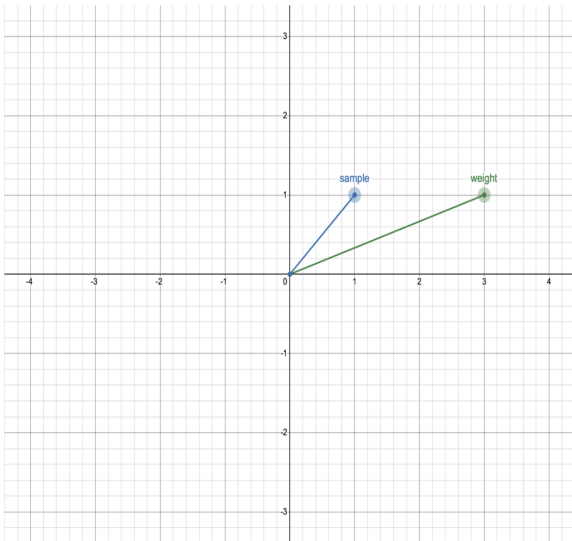
For this example, let's assume a starting weight vector of (3, 1). We'll also assume that the samples have the following vectors and labels:

- Sample₁ | (1, 1) | *Positive*
- Sample₂ | (-1, 1) | *Positive*
- Sample₃ | (2, -1) | *Negative*

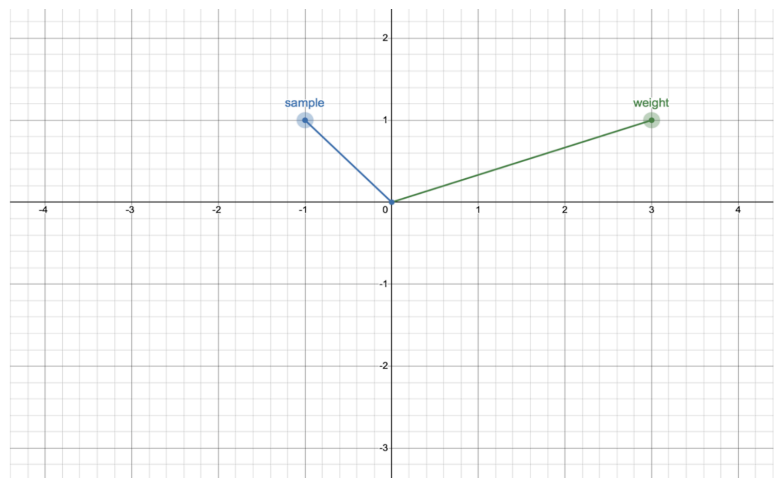
The binary perceptron training process is as follows:

1. Take the dot product of each sample with the weight vector.
2. Predict the label
 - *Positive* if value > 0
 - *Negative* if value ≤ 0
3. Update the weight vector
 - If the prediction is correct
 - No update needed
 - If the prediction is incorrect
 - If the prediction is a *False Negative* (true label being *Positive*)
 - Push the weight vector toward the sample vector
 - Add the sample vector to the weight vector
 - Similar samples will now lead to greater dot products
 - More likely to be classified as *Positive*
 - If the prediction is a *False Positive* (true label being *Negative*)
 - Pull the weight vector away from the sample vector
 - Subtract the sample vector from the weight vector
 - Similar samples will now lead to lesser dot products
 - More likely to be classified as *Negative*

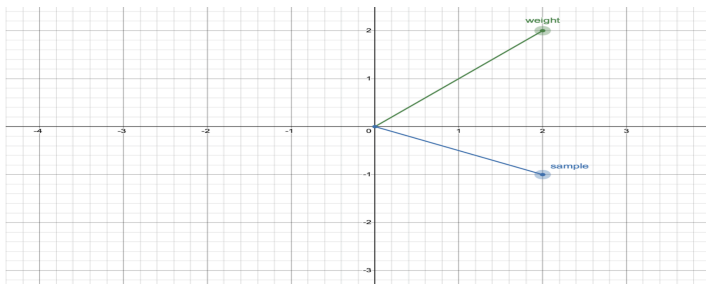
Ex. 1



- Dot Product:
 - sample · weight = (1, 1) · (3, 1) = 3 + 1 = 4
- Prediction:
 - *Positive (+)*
- Gold Label:
 - *Positive (+)*
- Correct:
 - Yes ☒
- Update:
 - No updates needed!



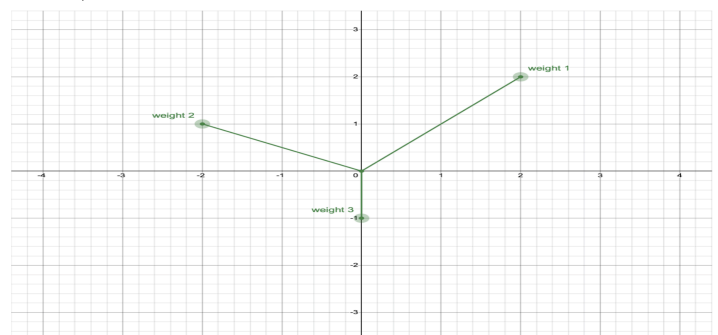
- Dot Product:
 - sample · weight = (-1, 1) · (3, 1) = -3 + 1 = -2
- Prediction:
 - *Negative (-)*
- Gold Label:
 - *Positive (+)*
- Correct:
 - No ☒
- Update:
 - *Add* — since the true label is *Positive*
 - weight_{new} = weight_{old} + sample = (3, 1) + (-1, 1) = (2, 2)
- See New Graph Below:



- Dot Product:
 - sample · weight = (2, -1) · (2, 2) = 4 - 2 = 2
- Prediction:
 - *Positive (+)*
- Gold Label:
 - *Negative (-)*
- Correct:
 - No ☒
- Update:
 - *Subtract* — since the true label is *Negative*
 - weight_{new} = weight_{old} - sample = (2, 2) - (2, -1) = (0, 3)

Multiclass perceptron algorithm question

Initial Setup:



For this example, let's assume starting weight vectors of (2, 2), (-2, 1), and (0, -1). We'll also assume that the samples have the following vectors and labels:

- Sample₁ | (-1, 0) | Class 3
- Sample₂ | (-1, 2) | Class 2

The multiclass perceptron training process is as follows:

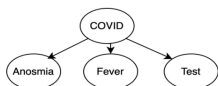
1. Take the dot product of each sample with each weight vector.
2. Predict the label
 - Argmax: predict label associated with the largest dot product value
3. Update the weight vectors
 - If the prediction is correct
 - No updates needed
 - If the prediction is incorrect
 - Update the incorrectly predicted class's weight vector
 - Pull this weight vector away from the sample vector
 - Subtract the sample vector from this weight vector
 - Similar samples will now lead to lesser dot products
 - Less likely to be predicted as this class
 - Update the true class's weight vector
 - Push this weight vector toward the sample vector
 - Add the sample vector to this weight vector
 - Similar samples will now lead to greater dot products
 - More likely to be predicted as this class

Example 2

- Dot Product:
 - sample · weight₁ = (-1, 2) · (2, 2) = -2 + 4 = 2
 - sample · weight₂ = (-1, 2) · (-1, 1) = 1 + 2 = 3
 - sample · weight₃ = (-1, 2) · (-1, -1) = 1 - 2 = -1
- Prediction:
 - Class 2
- Gold Label:
 - Class 2
- Correct:
 - Yes ✓
- Update:
 - No updates needed!

3. The following Bayes' net includes some of the symptoms COVID-19 and outcome of a COVID test. As shorthand, + and - are used to indicate an assignment for each binary random variable, C, A, F, T . The relevant marginal and conditional probabilities are provided in the accompanying table. Use the equation for joint distribution of a Bayes' net: $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$ to derive the probability that a patient presenting to the ER for some unrelated reason has COVID in absence of fever and anosmia and tests positive for COVID. (Author: Peyman)

- a. 0.1
- b. 0.01
- c. 0.0011
- d. 0.0001



$P(+C) = 0.005$
$P(+A +C) = 0.38$
$P(+F +C) = 0.48$
$P(+T +C) = 0.7$

▼ Answer and Explanation

Answer: c

Explanation:

$$\begin{aligned}
 P(+C, -A, -F, +T) &= \\
 P(+C) \cdot P(-A | +C) \cdot &P(-F | +C) \cdot P(+T | +C) \\
 &= \\
 (0.005)(0.62)(0.52)(0.7) &= \\
 0.0011
 \end{aligned}$$

- Dot Product:

- sample · weight₁ = (-1, 0) · (2, 2) = -2 + 0 = -2
- sample · weight₂ = (-1, 0) · (-2, 1) = 2 + 0 = 2
- sample · weight₃ = (-1, 0) · (0, -1) = 0 - 0 = 0

- Prediction:

- Class 2

- Gold Label:

- Class 3

- Correct:

- No ✗

- Update:

- Subtract the feature vector from the incorrectly guessed class's weight vector
 - weight_{new_2} = weight_{old_2} - sample = (-2, 1) - (-1, 0) = (-1, 1)
- Add the feature vector to the true class's weight vector
 - weight_{new_3} = weight_{old_3} + sample = (0, -1) + (-1, 0) = (-1, -1)

Example 1

1. Assume A is conditionally independent of B , given C ; i.e., $A \perp B | C$. Which of the following statements is not correct? (Author: Peyman)

- a. $P(A, B | C) = P(A | C) \cdot P(B | C)$
- b. $P(A, B | C) = P(A | B) \cdot P(B | C)$
- c. $P(A | B, C) = P(A | C)$
- d. $P(B | A, C) = P(B | C)$

▼ Answer and Explanation

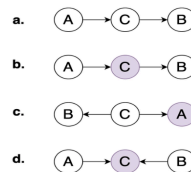
Answer: b

Explanation:

a and c are correct and are the two definitions of conditional independence.

d is correct because conditional independence works both ways for A and B: if $A \perp B | C$, then $B \perp A | C$. Therefore, $P(B | A, C) = P(B | C)$.

2. Which triple below guarantees that A is independent of B? Assume that there are no other paths between these nodes. Shading indicates that a node is observed (has a value assignment). (Author: Peyman)



▼ Answer and Explanation

Answer: b

4. What is NOT true of causality in the context of Bayes' nets? (Author: Peyman)

- a. It is possible to design a valid Bayes' net with the direction of causality reversed.
- b. Causality always exists in problems that are modeled as Bayes' nets and can be used to simplify the design of the net.
- c. The topology of the Bayes' net (the arcs and their direction) may or may not indicate causality.
- d. The topology of the Bayes' net always encodes the conditional independence assumptions.

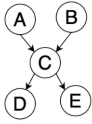
▼ Answer and Explanation

Answer: b

Explanation:

b is incorrect. Causal relationship may not even exist in real-world problems that are modeled as Bayes' nets. There may just be a correlation between the variables. The other answers are correct and are directly from the lecture.

5. Assuming all of the random variables are binary, what is the size of the following Bayes net (the number of probability table entries that fully describe it)? (Author: Peyman)
- 40
 - 32
 - 20
 - 16



▼ Answer and Explanation

Answer: c

Explanation:

In asymptotic notation, the size of a Bayes' net is $O(N \cdot 2^{k+1})$ where N is the number of nodes and k is the greatest number of parents for any node. In this case, it is possible to compute the size precisely. A and B have no parents, so they each have a marginal probability table of 2 entries. C has 2 parents, so its CPT has $2^{2+1} = 8$ entries. D and E each have one parents, so the size of their CPT is $2^{1+1} = 4$.

So, the size of the Bayes' net is $2 + 2 + 8 + 4 + 4 = 20$. By contrast the size of the joint probability distribution for these 5 binary variables would be $2^5 = 32$.

Answer: d

Explanation:

We are interested in the following joint probability: $P(-b, -e, +a, -j, -m)$ and will use the following formula:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

$$\begin{aligned}
 &P(-b, -e, +a, -j, -m) \\
 &= P(-b) \cdot P(-e) \cdot P(+a | -b, -e) \cdot P(-j | +a) \cdot P(-m | +a) \\
 &= 0.999 \times 0.998 \times 0.001 \times 0.1 \times 0.3 = 0.00003
 \end{aligned}$$

7. In the alarm Bayes' net in the previous question, what is the probability of an earthquake (with no burglary) that does NOT sound the alarm and Mary and John, being too worried about their own house, do NOT call? (Author: Peyman)

- 0.013
- 0.0013
- 0.00013
- 0.00001

▼ Answer and Explanation

Answer: b

Explanation:

$$\begin{aligned}
 &P(-b, +e, -a, -j, -m) \\
 &= P(-b) \cdot P(+e) \cdot P(-a | -b, +e) \cdot P(-j | -a) \cdot P(-m | -a) \\
 &= 0.999 \times 0.002 \times 0.71 \times 0.95 \times 0.99 = 0.0013
 \end{aligned}$$

16. The probability of every variable in a Bayes' net is conditioned only on its children. (Author: Peyman)

- True
- False

▼ Answer and Explanation

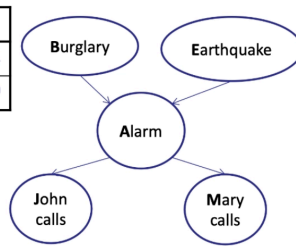
Answer: False

Explanation: The probability of every variable in a Bayes' net in conditioned only on its **parents**. This allows us eliminate all non-parent nodes from conditional probability tables (CPTs) of each node and leads to the equation:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

6. Given the Bayes' net below, what is the probability that a false alarm is sounded in absence of burglary or earthquake and John and Mary (perhaps jaded by too many false alarms) don't call? (Author: Peyman)
- 0.03
 - 0.003
 - 0.0003
 - 0.00003

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

14. What is NOT true of inference by enumeration when applied to Bayes' nets? (Author: Peyman)

- Works by categorizing variables into evidence, hidden and query variables, then summing (marginalizing) over the probabilities of hidden variables, then normalizing.
- It is an effective method of inference but is computationally costly and eliminates the compactness advantage of Bayes' nets.
- It involves reconstructing a large joint probability distribution from the Bayes' net which is exponential in the number of evidence variables.
- It can be used to answer what assignment to query variable Q is the most likely explanation (most consistent with) all the evidence variables?

▼ Answer and Explanation

Answer: c

15. Every variable in a Bayes' net is independent of all of its descendants given its children. (Author: Quiz)

- True
- False

▼ Answer and Explanation by JianFei

Answer: False

1

True or False 1 point Question

The Naive Bayes model is "naive" because it assumes that the features are conditionally independent of each other, given the class.

☒ True

☐ False

2

Multiple Choice 1 point Question

Write down the form of the joint probability model $P(X_1, X_2, X_3, Y)$ for this data using the Naive Bayes assumption. (Y is the class)

☐ $P(Y)P(X_1)P(X_2)P(X_3)$

☐ $P(Y)P(X_1|Y)P(X_2|X_1)P(X_3|X_2)$

☒ $P(Y)P(X_1|Y)P(X_2|Y)P(X_3|Y)$

☐ $P(Y)P(X_1|Y)P(X_2|X_1, Y)P(X_3|X_1, X_2, Y)$

3

True or False 1 point Question

In the perceptron training algorithm, weights are updated after every training instance.

☐ True

☒ False

4 Multiple Choice 1 point Question

Consider a two-dimensional data distribution where points belonging to class A are arranged around the origin in a circle with radius r_a , and points belonging to class B are arranged around the origin in a circle with radius r_b , where $r_a < r_b$.

When applied to this classification problem, the perceptron algorithm

- ☐ cannot be used to separate the two classes.
- ☒ can separate the two classes if we use feature augmentation.
- ☐ can separate the two classes if we decay the learning rate throughout training.
- ☐ can separate the two classes if we iterate through the instances in a suitable order.

5 Multiple Choice 1 point Question

For which of the following datasets would a single perceptron be unable to find a separating boundary (assuming we do not add additional features to the data):

- ☐ $Class A : (0, 0), (0, 1), (1, 0); Class B : (1, 1)$
- ☐ $Class A : (0, 0); Class B : (0, 1), (1, 0), (1, 1)$
- ☒ $Class A : \{(0, 0), (1, 1)\}; Class B : \{(0, 1), (1, 0)\}$
- ☐ $Class A : (0, 0), (0, 1), (1, 0), (1, 1); Class B :$

6 True or False 1 point Question

Given a non-random dataset, the perceptron algorithm guarantees that the weight vector will converge after a finite number of training iterations.

- ☐ True
- ☒ False

7 Multiple Choice 1 point Question

The inference procedure for the perceptron classification algorithm is best summarized as:

- ☒ Compute a weighted sum, then apply a threshold.
- ☐ Apply a threshold, and then compute a weighted sum.
- ☐ Add a bias value to each feature, then compute a weighted sum.
- ☐ Compute the magnitude of the weight vector, then of the input vector, and then threshold their sum.
- ☐ Try to do linear algebra, give up, and cast a "datum separatus" charm.

2. For supervised machine learning, we divide the data into 3 sets. Which statement is NOT true? (Author: Peyman)

- a. Training set is used to derive model parameters
- b. Held-out or development (dev) or validation set is used to tune the hyper-parameters
- c. Test set is used to assess the model's accuracy.
- d. The three sets need not be disjoint

▼ Answer and Explanation

Answer: d

Explanation:

The 3 sets need to be disjoint in order to allow the model to generalize to new inputs. When there is not enough data to split into 3 disjoint sets, the book (Section 19.4) describes a workaround called k-fold-cross-validation to have each example serve double duty — as training data and validation data — but **not at the same time**.

3. Which statement is correct about bias and variance? (Author: Peyman)

- a. Variance is the tendency of a model's predictions to deviate from the expected values when averaged over different training sets.
- b. Bias is the amount of change in the model due to fluctuations in the training data.
- c. high bias leads to over-fitting.
- d. high variance leads to over-fitting.

▼ Answer and Explanation

Answer: d

1. What is NOT true of supervised model-based machine learning? (Author: Peyman)

- a. Given a training set of n input-output pairs (x_i, y_i) , generated by an unknown function $y = f(x)$, finds a function h that approximates the true function f .
- b. It learns a model (hypothesis) and uses it to predict an output $\hat{y}$ for any future input x
- c. It does not require labeled data
- d. It can be used for both classification (categorical output variables) and regression (numerical output variables) tasks

▼ Answer and Explanation

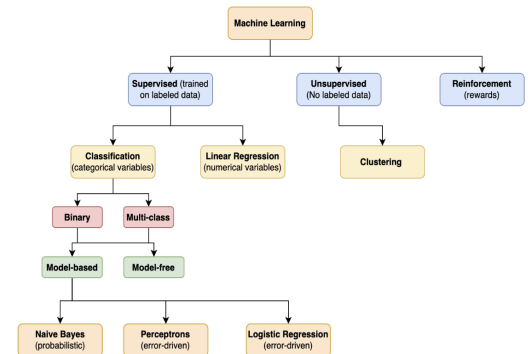
Answer: c

Explanation:

a and b are true (Russell and Norvig 19.2).

c is incorrect. By definition, supervised learning required labeled data

d is true. See the hierarchy below:



4. Label each statement below about naive Bayes and perceptrons as true or false.

(Author: Peyman)

- Naive Bayes and Perceptrons are both model-based classifiers.
- Naive Bayes uses a linear model that is refined by an error-driven process.
- Perceptrons use a probabilistic model.
- Naive Bayes is used for binary and multi-class classification whereas perceptrons are only used for binary classification.
- Naive Bayes uses one pass over the data to count and compute probabilities
- Perceptrons make multiple passes through the data to improve the weight vector(s) and decision boundary
- Empirically, perceptrons are often more accurate than naive Bayes.

▼ Answer and Explanation

Answer: a: true, b: false, c: false, d: false, e: true, f: true, g: true

Explanation:

a is true. Naive Bayes uses a probabilistic model based on a Bayes' net. Perceptrons use a linear model $h = \mathbf{w} \cdot \mathbf{f}(\mathbf{x}) + b$

b and c are false. The descriptions are switched.

d is false. Both can be used for both binary and multi-class classification

e and f are true. Self-explanatory.

g is true according to the lecture

5. What is NOT true of naive Bayes? (Author: Peyman)

- All features are independent of each other and only dependent on the label
- Generally, the model needs to learn not only the conditional probability for each feature $P(F_i|Y)$, but also the prior probability over labels $P(Y)$
- The number of parameters is exponential in the number of features
- Because of major independence assumption, the naive Bayes model is very simplistic but often works anyway.

▼ Answer and Explanation

Answer: c

Explanation:

a is true. All features are conditionally independent of each other given the label. This is the principle independence assumption in a naive Bayes model and is what makes it naive.

b is true. In some cases, where the class label is uniformly distributed, $P(Y)$ can be readily determined (e.g. in digit recognition, $P(Y) = 0.1$ for all class labels), but in general, it needs to be learned along with the conditional probabilities.

c is not true. The number of parameters is **linear** in the number of features

$$\theta = |Y| \cdot |F|^n$$

This is a consequence of the NB independence assumptions. Without them the number of parameters would be the size of the full joint distribution table $\theta = |Y| \cdot |F|^n$

d is stated in the lecture

7. A sequence of outcomes in 10 trials for a random variable X is

$(r, b, g, r, r, b, g, b, r, b)$. Select all computations that are correct. (Author: Peyman)

- $P_{ML}(r) = 4$
- $P_{ML}(w) = 0$
- $P_{LAP,2}(g) = \frac{1}{4}$
- $P_{LAP,2}(w) = 0$

▼ Answer and Explanation

Answer: b, c

Explanation:

$$P_{LAP,k}(x) = \frac{\text{count}(x) + k}{N + k \cdot |X|}$$

Where N = sample size, and $|X|$ = count of unique values in the sample

$$P_{ML}(r) = \frac{4}{10}$$

$$P_{ML}(w) = \frac{0}{10} = 0 \text{ since } w \text{ is not observed in the sequence}$$

$$P_{LAP,2}(g) = \frac{2+2}{10+(2)(3)} = \frac{1}{4}$$

$$P_{LAP,2}(w) = \frac{0+2}{10+(2)(3)} = \frac{1}{8}$$

6. What is NOT true of inference for naive Bayes. (Author: Peyman)

- It can use any standard inference technique for Bayesian networks such as inference by enumeration or variable elimination.
- It computes the conditional probability of the class label given the evidence
- Although it can compute the probability of a class label given an input, it can not classify that input.
- If inference is done by enumeration, it reconstructs the joint probability table, which is computationally expensive

▼ Answer and Explanation

Answer: c

Explanation:

a is true from the lectures.

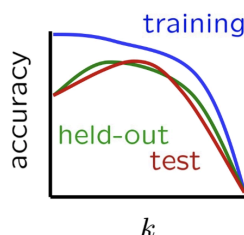
b is true. The goal of inference is to get $P(Y|F1...Fn)$

c is not true. After computing $P(Y|F1...Fn)$, we can take its $\arg \max_y$ to find the class label with the highest probability and classify the input with that label.

d is true. This is the main draw-back of inference by enumeration vs. the more sophisticated variable elimination inference algorithm.

8. Which equation most readily explains why accuracy over the training set monotonically decreases in the graph below as the hyper-parameter k increases during the fine-tuning process? (Author: Peyman)

- $P_{ML}(x) = \frac{\text{count}(x)}{\text{total samples}}$
- $P_{LAP,1}(x) = \frac{\text{count}(x)+1}{N+|X|}$
- $P_{LAP,0}(x) = P_{ML}(x)$
- $P_{LAP,1}(x) = P_{ML}(x)$



▼ Answer and Explanation

Answer: c

15. Why would a data set not be linearly separable? Select all that apply. (Author: Peyman)

- Examples may be labeled incorrectly, thus creating noise.
- We may not have enough features to fully expose the difference between classes.
- The classes may intrinsically overlap.
- Because the separator that correctly classifies all data points is non-linear.

▼ Answer and Explanation

Answer: all are correct

16. What is true of MIRA and support vector machines (SVM)? Se

(Author: Peyman)

- a. Both are variants of perceptrons.
- b. Both provide better generalizability than a regular perceptrc
- c. When $y \neq y^*$, MIRA moves the weight vectors more than a
- d. SVM treats all data points equally.

▼ Answer and Explanation

Answer: a, b

3 True or False 1 point Question

For linearly separable data, there exists only one decision boundary that separates the classes.

- ☐ True
- ☒ False

4 Multiple Choice 1 point Question

Is it possible that the assignment of observations to clusters do not change between successive iterations in K-Means?

- ☒ Yes
- ☐ No

5 Multiple Choice 1 point Question

Relative to single-layer perceptrons, neural networks gain their power from

- ☐ smooth activation function
- ☒ stacking of layers
- ☐ convolution

1 Multiple Answer 1 point Question

Which of the following are true about linear classifiers? Please check all that apply.

- ☒ They can be used for both regression and classification.
- ☐ All sets of data points are linearly separable.
- ☒ When using the perceptron learning rule, the weights are updated when the actual output does not match the hypothesis output.
- ☐ The learning rule must be applied to one example at a time.

2 Multiple Choice 1 point Question

Which of the following best characterizes the difference between parametric and nonparametric models?

- ☐ Parametric models cannot summarize data with a large number of training examples.
- ☐ Parametric models can be used if each hypothesis considers all of the other training examples to make the next prediction.
- ☐ Instance-based learning and memory-based learning use parametric models.
- ☒ A parametric model has a fixed size on the number of parameters.

6 Multiple Answer 1 point Question

In neural networks, nonlinear activation functions: Please check all that apply.

- ☒ Make it possible to do the gradient calculation in backpropagation, as opposed to using step function which isn't differentiable.
- ☒ Help to learn nonlinear decision boundaries.
- ☐ Are applied only to the output units.
- ☐ Always output values between 0 and 1.

7 True or False 1 point Question 7

When performing k-means clustering, each observation always starts in its own cluster, and then pairs o

- ☐ True
- ☒ False

1. What is true of both logistic regression and perceptrons? Select all that apply.

(Author: Peyman)

- a. They are both model-based classifiers.
- b. They both compute $z = \mathbf{w} \cdot \mathbf{x} + b$
- c. They both directly compare z against a threshold to classify the input.
- d. They both use a loss function to learn from training examples.

▼ Answer and Explanation

Answer: a, b

2. What is true of the loss function for logistic regression? (Author: Peyman)

- a. Either mean squared error or cross entropy can be used as the loss function for logistic regression
- b. Mean squared error is a convex function for both linear regression and logistic regression.
- c. Cross entropy has only one global minimum and no other local minima; therefore, it is a suitable function for stochastic gradient descent.
- d. Cross entropy is not a convex function for logistic regression.

▼ Answer and Explanation

Answer: c

3. Which statement is correct about cross entropy loss? (Author: Peyman)

- a. When the actual label $y = 0$, then $L_{CE} = -\log \sigma(z)$
- b. When the actual label $y = 1$, then $L_{CE} = -\log(1 - \sigma(z))$
- c. L_{CE} can be negative.
- d. L_{CE} is small if the model's estimate is close to correct and is larger if the estimate deviates from correct.

► Answer and Explanation

4. What is NOT true of stochastic gradient descent? (Author: Peyman)

- a. Minimizes the loss function
- b. Processes the input and modifies the model parameters example by example rather than waiting to see the entire input
- c. Is an off-line algorithm
- d. Nudges θ in the direction opposite that of the gradient

▼ Answer and Explanation

Answer: c

Explanation:

- Stochastic gradient descent is an **online** algorithm that minimizes the loss function by **computing its gradient after each training example**, and **nudging θ** in the right direction (the opposite direction of the gradient).
- An **online algorithm** is one that processes its input **example by example**, rather than waiting until it sees the entire input.

9. L1 and L2 regularization are two methods to address the problem of overfitting in logistic regression. Match each characteristic below with L1, L2 or both. (Author: Peyman)

- a. Lasso
- b. Ridge
- c. Euclidean distance
- d. Manhattan distance
- e. Linear function of weight values
- f. Quadratic function of weight values
- g. Produces a regularization term $R(\theta)$ that is added to the loss function
- h. Penalizes weight vectors with too many large weights

▼ Answer and Explanation

Answer: a: L1, b: L2, c: L2, d: L1, e: L1, f: L2, g: both, h: both

5. What is NOT true of the gradient in gradient descent. (Author: Peyman)

- a. Consists of the partial derivatives of the loss function with respect to each one of the model parameters.
- b. In a multi-dimensional feature space, the gradient is a single slope (derivative).
- c. The gradient is the difference between the true output and the estimated output multiplied by the individual feature value.
- d. When the gradient with respect to a particular weight is negative, that weight is increased by the algorithm.

▼ Answer and Explanation

Answer: b

11. The input layer of a neural network (Author: Peyman)

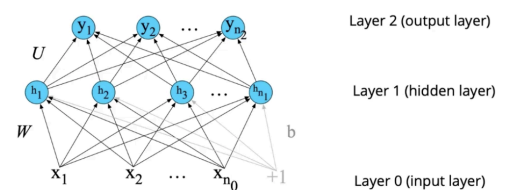
- a. is called layer 1.
- b. consists of logistic regression units.
- c. is of a vector of features.
- d. has the same number of units as the first hidden layer so that there is a one-to-one mapping between the two layers.

▼ Answer and Explanation

Answer: c

Explanation:

The input layer is called layer 0 because it has no neurons (logistic regression units). Rather the nodes(units) are just the individual features of the input example (including the bias feature), as shown below. So the input layer is the feature vector. There is no one-to-one mapping between the number of input features and the number of neurons in the first hidden layer, since each feature is fed to every neuron in layer 1.



10. A neural network is a network of computational units (neurons) in which: (Author: Peyman)

- a. Every neuron is connected to every other neuron.
- b. Each neuron (unit) receives its input from one neuron in the preceding layer.
- c. Each unit receives its inputs from all units in the preceding layer.
- d. Each unit receives its inputs from some units in the preceding layer as specified by the neural network architecture.

▼ Answer and Explanation

Answer: c

12. What is true of the activation function g for neurons in a neural network? Select all that apply. (Author: Peyman)

- a. It is the same as the sigmoid function
- b. It takes $z = \mathbf{w} \cdot \mathbf{x} + b$ as its argument
- c. It returns a value that is called the activation value (or activation) of that neuron
- d. All neurons in the network must have the same activation function.

▼ Answer and Explanation

Answer: b, c

Explanation:

- a is false. The activation function may or may not be the sigmoid function. Other activation functions include ReLU (rectified linear unit) and tanh functions.
- b and c are correct. The activation function acts upon $z = \mathbf{w} \cdot \mathbf{x} + b$ to produce an activation value (or just activation) a of that neuron, so $a = g(\mathbf{w} \cdot \mathbf{x} + b)$. Thus the activation of a neuron is the output of that neuron, which serves as one of the inputs to every neuron in the next layer.
- d is false. Neurons in different layers may be assigned different activation functions, and there are good reasons for doing so. PyTorch and TensorFlow both allow you to do this.

13. What is true of activation in a neural network? (Author: Peyman)

- a. The activation (output) of a neuron is a scalar value.
- b. Each layer has an activation.
- c. The activations of all neurons in one layer become the features for each neuron in the next layer.
- d. All of the above

▼ Answer and Explanation

Answer: d

14. A perceptron can be thought of as having an activation function that is a step function. A perceptron is not a good computational unit for a neural network compared to a sigmoid neuron. Select all the reasons that apply. (Author: Peyman)

- a. A step function is not a smooth function.
- b. A small change in weights and bias can lead to a large change in output of the perceptron, an undesirable property for a neural network.
- c. Neural networks are binary classifiers and do not need the multi-classification capability of the perceptron.
- d. The step activation function of a perceptron does not lend itself to differentiation, which would create a problem for learning and back-propagation.

▼ Answer and Explanation

Answer: a, b, d

17. What is true of error back propagation? Select all that apply. (Author: Peyman)

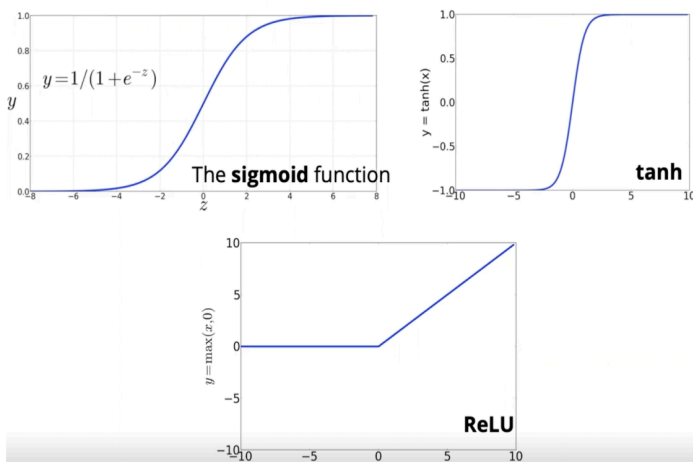
- a. It is used when a trained neural network is used to categorize a sample with a set of features.
- b. It is required for training of a neural network.
- c. It uses the principle of backward differentiation in a computation graph.
- d. It uses the chain rule to compute the derivative of the loss function with respect to the weights for each neuron in the network.

▼ Answer and Explanation

Answer: b, c, d

Explanation:

In order for a neural network to act as binary classifier, its final output must be a probability between 0 and 1. Sigmoid and tanh activation functions produce such an output, but ReLU does not. The output of ReLU is linear for $z \geq 1$ and its range is $[0, +\infty]$.



1 Multiple Choice 1 point Question

Which of the following is not a common benefit of using log probabilities instead of probabilities with a Categorical distribution?

- ☐ Underflow results from dealing with small floating point numbers, which is avoided using log probabilities.
- ☒ A log probability representation of a categorical distribution is more efficient to sample.
- ☐ In certain instances, it's easier to optimize over sums than products.

2 True or False 1 point Question

Markov models are stationary because probabilities assigned to a sequence are invariant with respect to shifts in data [train, validation, test, etc.]

- ☐ True
- ☒ False

3

Multiple Choice 1 point Question

Cross Entropy:

- ☐ is a lower bound on the entropy. The more accurate a model is, the lower the cross entropy.
- ☒ is an upper bound on the entropy. The more accurate a model is, the lower the cross entropy.
- ☐ is a lower bound on the entropy. The more accurate a model is, the higher the cross entropy.
- ☐ is an upper bound on the entropy. The more accurate a model is, the higher the cross entropy.

4

Multiple Choice 1 point Question

Vector representations of words using term-document matrices and word-word matrices can account for polysemy.

- ☐ True
- ☒ False
- ☐ Not enough information

5

Multiple Choice 1 point Question

How many tokens would the following sentence be broken into:

ENIAC's construction was financed by the United States Army, Ordnance Corps, led by Major General Gladeon M. Barnes.

- ☐ 21
- ☐ 22
- ☐ 23
- ☒ 24

6

Multiple Choice 1 point Question

Suppose you are building an n-gram model for a corpus with a vocabulary of size V and a total of T tokens. The running time to build this model depends on

- ☐ T
- ☐ n and V
- ☒ n and T
- ☐ n, T, and V

▼ Module 13 Questions: Natural Language Processing

1. The distributional hypothesis states that: (Author: Peyman)

- a. Words that occur in similar contexts tend to have similar meanings.
- b. Words maintain a constant meaning regardless of their context.
- c. Words in similar syntactic (grammatical) structures have similar meanings.

▼ Answer and Explanation

Answer: a**Explanation:**

a is the definition of distributional hypothesis (Chapter 6)

b is incorrect. First, some words are polysemous and have different meanings (sense) in different contexts. Second, the distributional hypothesis posits that the meaning of a word can be derived from its context (even for words that have one sense), which is inconsistent with the statement in b.

c is incorrect. the distributional hypothesis focuses on semantic context, not just syntactic structure.

2. Which statement is correct? (Author: Peyman)

- a. Synonymy is the same as word similarity.
- b. Word similarity is the same as word relatedness.
- c. A semantic frame is a set of words that denote perspectives or participants in a particular type of event.
- d. Connotation is the affective meaning of the word and is measured along two dimensions: valence and arousal.

▼ Answer and Explanation

Answer: c

3. What is NOT true of vector semantics? (Author: Peyman)

- a. Is the standard way to represent word meaning in NLP.
- b. Represents a word as a point in a multidimensional semantic space that is derived from the distributions of word neighbors.
- c. Word vectors have the same length and differ only in their direction.
- d. Defines the meaning of a word by its **distribution** in language use, meaning its neighboring words or grammatical environments.

▼ Answer and Explanation

Answer: c**Explanation:**

c is incorrect. Word vectors have different lengths. More frequently occurring words have longer vectors. This creates a need for normalization of the vector dot product with respect to word lengths when computing word similarity.

4. Word similarity is best measured by: (Author: Peyman)

- a. Euclidean distance between two words across all N dimensions: $\sqrt{\sum_{i=1}^N (v_i - w_i)^2}$. The shorter the distance, the more similar are the words.
- b. The dot product of the two word vectors: $\mathbf{v} \cdot \mathbf{w}$
- c. The dot product of the two word vectors, each normalized by division by its length $\frac{\mathbf{v}}{|\mathbf{v}|} \cdot \frac{\mathbf{w}}{|\mathbf{w}|}$
- d. The cosine of the angle between the two vectors, which is different from the dot product of two unit vectors

▼ Answer and Explanation

Answer: c

5. Which statement is true of co-occurrence matrices? (Author: Peyman)
- A term-document matrix has dimensions $|V| \times |V|$, where V is the vocabulary
 - A term-context matrix has dimensions $|V| \times D$, where D is the number of documents.
 - In a term-context matrix, each cell records the number of times the row (target) word and the column (context) word co-occur in some context in some training corpus.
 - In a term-document matrix, each row represents a document and each column represents a word.

▼ Answer and Explanation

Answer: c

Explanation:

a and b are incorrect. The answers are reversed.

c is correct. Each cell records the number of times the row (target) word and the column (context) word co-occur in some context in some training corpus. The context is a window around the word, e.g., 4 words to the left and 4 words to the right.

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
information	0	...	3325	3982	378	5	13	...

d is incorrect. In a term-document matrix, each column represents a document and each row represents a word:

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Figure 6.5 The term-document matrix for four words in four Shakespeare plays. The red boxes show that each word is represented as a row vector of length four.

7. What is NOT true of Word2Vec algorithms? (Author: Peyman)

- Word2Vec embeddings are static embeddings, meaning that the method learns one fixed embedding for each word in the vocabulary.
- Word2Vec trains a logistic regression classifier to compute the probability that two words are likely to occur nearby in text.
- The output of a Word2Vec algorithm is the prediction made by the classifier in b.
- The output of a Word2Vec algorithm is the set of weights that are computed by gradient descent in the process of training the classifier.

▼ Answer and Explanation

Answer: c

8. The parallelogram model is used for solving simple analogy problems of the form: *a is to b as a* is to what?* What is the correct equation for the parallelogram model? (Author: Peyman)

- $\mathbf{b}^* = \min(\text{distance}(\mathbf{x}, \mathbf{b} - \mathbf{a} + \mathbf{a}^*))$
- $\mathbf{b}^* = \arg \min_{\mathbf{x}} \text{distance}(\mathbf{x}, \mathbf{b} - \mathbf{a} + \mathbf{a}^*)$
- $\mathbf{b}^* = \text{distance}(\mathbf{b} - \mathbf{a}, \mathbf{a}^*)$
- $\mathbf{b}^* = \text{distance}(\mathbf{b} - \mathbf{a}^*, \mathbf{a})$

▼ Answer and Explanation

Answer: b

Explanation: We first compute the term $\mathbf{b} - \mathbf{a} + \mathbf{a}^*$, which puts us at a point in vector space close to $\mathbf{b}^*$. Then we look at all words $\mathbf{x}$ in the same vector space and select the word that has the minimum distance to the point $\mathbf{b} - \mathbf{a} + \mathbf{a}^*$. That word will be $\mathbf{b}^*$.

6. What is NOT true of sparse and dense word semantic models? (Author: Peyman)

- Sparse vectors are long and have many zeros.
- In sparse models, raw word frequencies are a good measure of association between words.
- Dense vectors are short and have few zeros.
- Regardless of whether a sparse or dense vector representation is used, word similarity is measured by cosine of the angle between the vectors.

▼ Answer and Explanation

Answer: b

10. The n-gram language model compared to neural language models: (Author: Richard)

- Both can generalize over similar words.
- N-gram is faster to train than Neural language models.
- N-gram LM can handle longer histories.
- Neural LM also need smoothing for unseen token context pair.

▼ Answer and Explanation

Answer: b

Explanation: one of disadvantage of neural LM is slower to train than n-gram LM

521_13.1_Neural Language models Neural Language Models

Neural Language models have several advantages over n-gram LMs:

- They don't need smoothing
- They can handle much longer histories.
- They can generalize over contexts of similar words.
- Neural LMs tend to have much higher predictive accuracy than n-gram LMs.

Disadvantage: slower to train than traditional n-gram LMs

like n-gram language models, neural language models can generalize over contexts of similar words. They achieve this by: (Author: Peyman)

- representing words by their embeddings, rather than just by their word identity.
- applying Laplace smoothing.
- using interpolation
- using backoff

Answer and Explanation

Answer: a

12. Ever since the advent of neural language models, n-gram language models have become obsolete and are only of historical and educational interest. (Author: Peyman)

- a. True
- b. False

▼ Answer and Explanation

Answer: b

Explanation:

The statement is false. From Jurafsky and Martin, Section 7.5: "[compared to n-gram LMs] neural net language models are much more complex, are slower and need more energy to train, and are less interpretable than n-gram models, so for many (especially smaller) tasks an n-gram language model is still the right tool."

13. What is NOT true of one-hot vector representation of a word in a neural LM?

(Author: Peyman)

- a. A one-hot vector is a bit vector of length $|V|$
- b. Only the bit representing the index of the input word in the vocabulary is 1. The other bits are 0.
- c. The one-hot vectors are used directly as the input layer to the feed forward network and are multiplied by the weight matrix $\mathbf{W}$ to provide the input to the first hidden layer.
- d. If the input to the neural network is 3 words and the output is the probability distribution of the 4th word, then there are 3 one-hot vectors in the model.

▼ Answer and Explanation

14. What is NOT true of the embedding matrix $\mathbf{E}$ in neural language models? (Author: Peyman)

- a. It has the size $d \times |V|$
- b. A one-hot vector representation of a word is multiplied by $\mathbf{E}$ to produce an embedding for that word.
- c. Either the embedding matrix can be provided to the language model and frozen throughout the training process or it can be learned during the training process along with the weights and biases for each layer.
- d. There is a separate embedding matrix for each input word. For example, if 3 words are used by the neural language model to predict the 4th word, then 3 different embedding matrices are learned during the training process.

▼ Answer and Explanation

Answer: d

15. What is true of the embedding layer $\mathbf{e}$ in a neural LM? (Author: Peyman)

- a. It is the concatenation of N individual embeddings representing the N input words.
- b. It is the vector sum of N individual embeddings representing the N input words.
- c. It is a matrix of dimensions $N \times V$, where each column represents an embedding of the N different input words.

▼ Answer and Explanation

Answer: a

Explanation:

The embedding layer is the **concatenation** of the embeddings for the input words. If there are 3 input words, the embedding layer is vector of length $3|V|$. This vector $\mathbf{e}$ takes the place of the feature vector $\mathbf{x}$, which is the input to a generic neural network.

16. What is true of training a feed-forward neural language model? (Author: Peyman)

- a. It is self-supervised in the sense that there are no human generated labels for each training example.
- b. It uses the cross entropy loss function and back propagation to learn the model parameters.
- c. It can learn not only the weights and biases for each layer in the network but also the embedding
- d. All of the above

▼ Answer and Explanation